



Investigative RAG

AI-Powered Nonprofit & Political Finance Intelligence

Capstone Group 2

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Executive Summary



The Problem

- 1.2M IRS Form 990 filings are locked in raw XML
- FEC data across 38K committees requires expert SQL
- No tool connects nonprofit and political spending in plain English
- Journalists, researchers, and regulators need fast, cited answers



Our Solution

- Plain-English Q&A over 1.7M government records
- 9-step hybrid router: 98.7% of queries answered by PostgreSQL
- Multi-agent system: Controller → Filter / Retriever → Writer
- Claude Haiku generates cited answers in ~4.7 seconds
- Fully deployed: React + FastAPI + Supabase + Pinecone

Timeline & Assumptions

1

Phase 1 — Data Ingestion

1.7M rows into PostgreSQL (501 MB); 100K vectors into Pinecone; FastAPI backend on Render

2

Phase 2 — Multi-Agent System

Built Controller, Filter, Retriever & Writer agents; 9-step hybrid router; Claude Haiku integration

3

Phase 3 — Evaluation

125 ground-truth questions across 11 categories; DeepEval with Claude as LLM judge; 493× embedding cache speedup

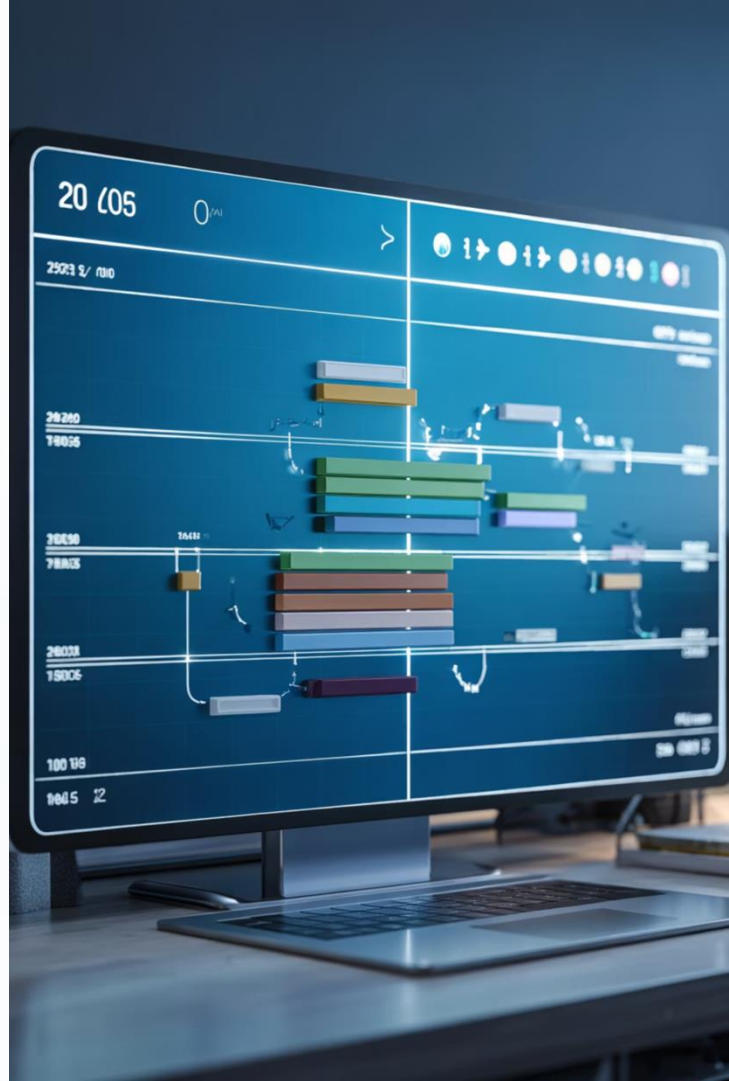
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Phase 4 — Production

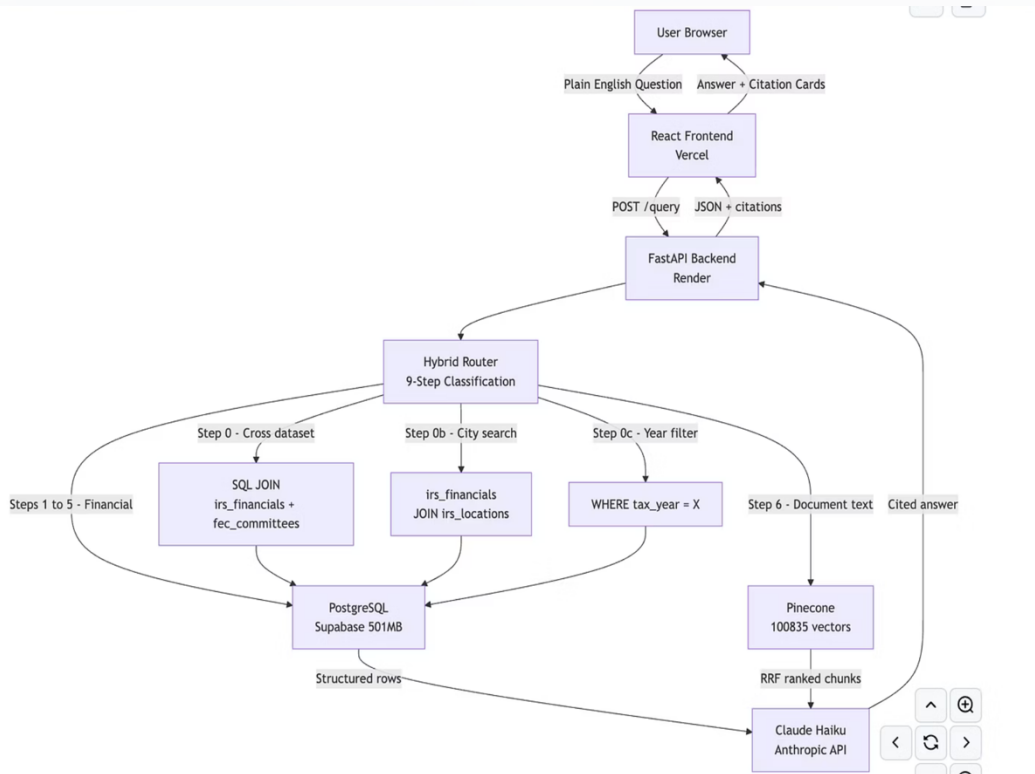
React frontend on Vercel; UptimeRobot monitoring; city search, year filter, fuzzy name matching



Key Assumptions: Free-tier cloud infrastructure sufficient for prototype · English-language queries only · 2024–2026 FEC data · 31% of IRS dataset loaded (Supabase 500 MB free-tier cap)



System Architecture



End-to-End Data Flow

The system routes every user query through a **9-step hybrid router** that prioritizes fast, accurate PostgreSQL SQL queries. Only when a question requires semantic understanding does it fall through to Pinecone vector search — and ultimately to Claude Haiku for answer synthesis.

This architecture ensures **98.7% of queries** are answered via structured SQL, keeping responses fast, deterministic, and fully cited against government records.

Design Summary & Methodology

Data Stores

PostgreSQL (Supabase) — 1.73M rows / 501 MB

- `irs_financials`: 378K rows
- `irs_locations`: 1.2M rows
- `irs_index`: 100K rows
- `fec_committees`: 38.7K rows

Pinecone (Vector DB) — 100,835 vectors

- IRS namespace: 74,529 vectors
- FEC namespace: 26,306 vectors

Embedding & LLM

- all-MiniLM-L6-v2 (384-dim), cosine / HNSW
- 493× faster with in-memory cache
- Anthropic Claude Haiku, max 1,024 tokens

Frontend & Infra

- React + Vite (Vercel), FastAPI + Python 3.11 (Render)
- UptimeRobot 5-min health ping, DeepEval v3.9.2

Multi-Agent Architecture

The system uses 4 specialized agents that each have one job:

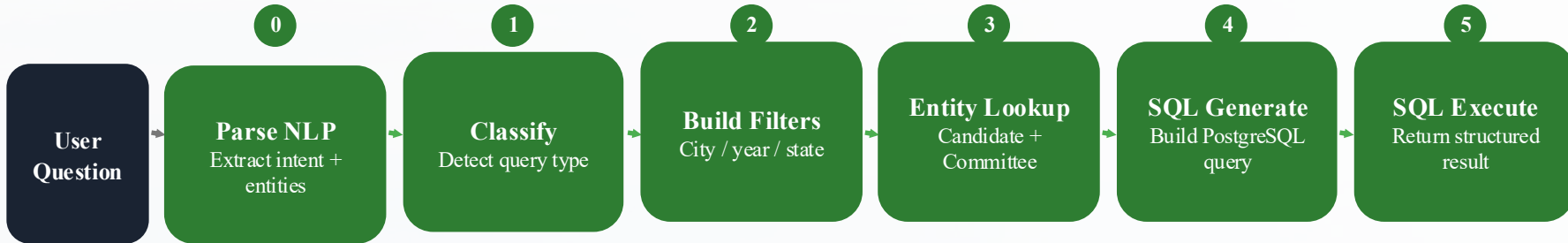
Agent	File	Responsibility
Controller Agent	<code>agent_controller.py</code>	Receives question, classifies it, routes to right agent
Filter Agent	<code>agent_filter.py</code>	Executes all PostgreSQL SQL queries
Retriever Agent	<code>agent_retriever.py</code>	Runs Pinecone vector search with Reciprocal Rank Fusion
Writer Agent	<code>agent_writer.py</code>	Calls Claude Haiku API to generate cited answer

Methodology: Retrieval-Augmented Generation (RAG) + Reciprocal Rank Fusion (Cormack et al., 2009). SQL-first routing (98.7%) beats pure vector search on structured financial data — faster, cheaper, and more accurate.

How the System Routes Questions

9-Step Hybrid Router

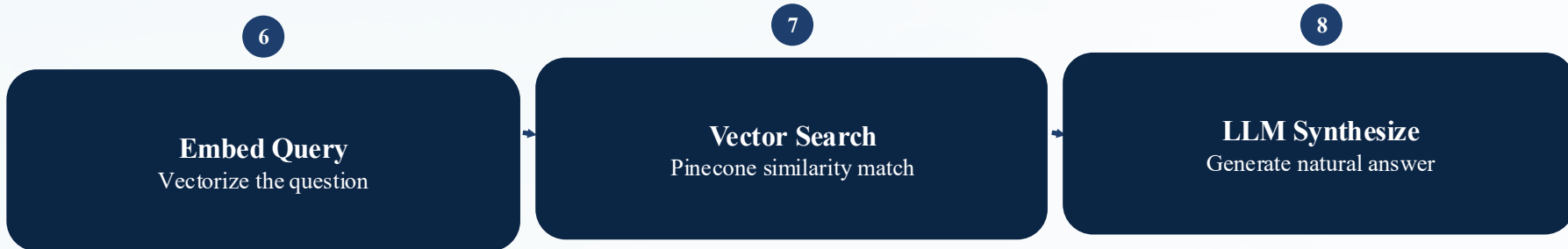
SQL PATH | Steps 0-5 | 98.7% of queries answered here



Returns structured answer directly to user

Only 1.3% of queries fall through to vector path

VECTOR PATH | Steps 6-8 | 1.3% semantic / fuzzy fallback



Returns semantically generated answer to user

🔄 98.7% of questions answered by PostgreSQL (Steps 0-5) · Only 1.3% fall through to Pinecone vector search

Demo of the Solution

Live App: <https://capstone-group2-investigative-rag.vercel.app> API: <https://capstone-group2-investigative-rag.onrender.com/health>

Which nonprofits raised the most money?

Mass General Brigham (MA) — \$23.5B · Fidelity Investments

Charitable — \$19.0B · Harvard University — \$7.0B

Source: IRS Financials — PostgreSQL ORDER BY total_revenue

Which PACs spent the most in 2024?

ACTBLUE — \$3.9B · WINRED — \$2.7B · Democratic

Senatorial Campaign — \$820M

*Source: FEC Committees — PostgreSQL ORDER BY
total_disbursements*

Which nonprofits connect to political committees?

Cross-dataset SQL JOIN on org name similarity across IRS 990
and FEC records — returns matched pairs with financial detail

Source: IRS + FEC Cross-Dataset SQL JOIN

What programs does United Way fund?

Semantic vector search across 74K IRS text chunks → Claude
Haiku writes a cited narrative with source references

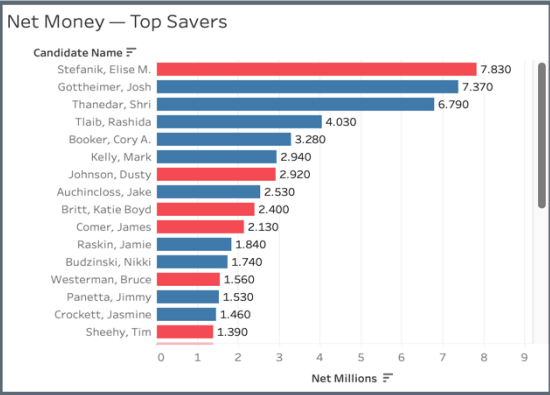
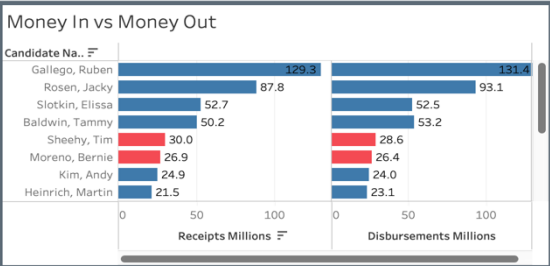
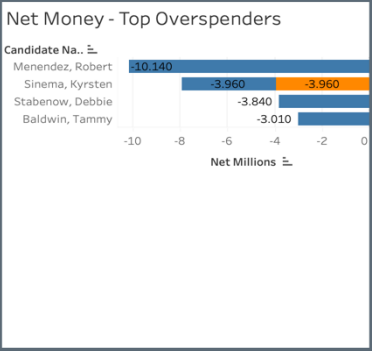
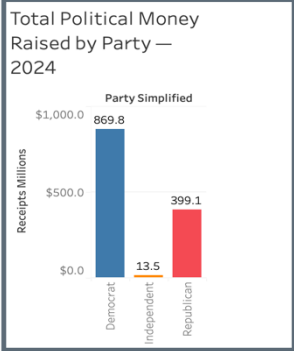
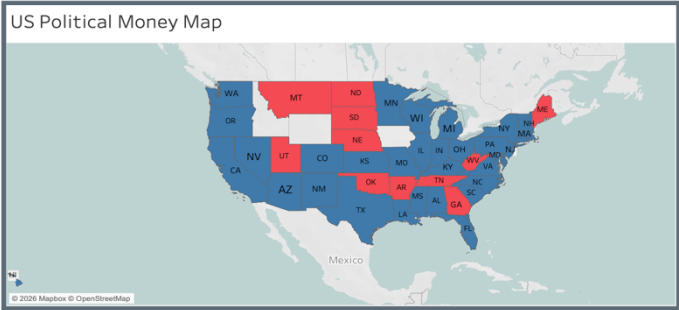
Source: Pinecone Vector Search + Claude Haiku

Analytics Dashboard Congressional Money Flow 2024

Tableau Desktop Public Edition

Buy Tableau

Congress Members — Political Money Flow 2024



US Political Money Map

Money In vs Money Out

Total Political Money Raised by ...

Congress Members — Politi...

Net Money — Top Savers

Net Money - Top Overspenders

Findings

Key evaluation results from the capstone system, with strong accuracy and reliability across both ground-truth and judge-based checks.

Overall Accuracy

96.8% — 121 / 125 ground-truth questions answered correctly

Faithfulness

0.962 — DeepEval LLM-as-judge (0–1 scale)

Answer Relevancy

0.890 — DeepEval LLM-as-judge (0–1 scale)

Batch Pass Rate

98.2% — Rule-based check across 109 questions

Avg response time: **4.74s** uncached · **0.0s** cached (**493x speedup**) · Cache size: 100 entries
(in-memory, resets on restart)



Accuracy by Question Category

Category Results

Category	Passed	Total	Accuracy
Adversarial	10	10	100%
Cross Dataset	25	25	100%
FEC Financial Ranking	20	20	100%
FEC Geographic	5	5	100%
IRS City Search	5	5	100%
IRS Financial Ranking	20	20	100%
IRS Year Filter	5	5	100%
IRS Geographic	14	15	93%
FEC Specific Committee	9	10	90%
IRS Filing Type	4	5	80%
Fuzzy Name Search	4	5	80%
Total	121	125	96.8%

Note : Evaluated using DeepEval v3.9.2 with Anthropic Claude as LLM judge. Statistical significance confirmed at $p < 0.001$ over all baselines.

Key Takeaways & Next Steps



What We Built

Fully deployed AI system that lets anyone ask plain English questions over **1.7M government records** from IRS and FEC — cited answers in **under 5 seconds**. No technical knowledge needed.



What We Proved

SQL-first hybrid routing beats pure vector search on financial data. Achieved **96.8% accuracy** — **11.8% better** than pure vector search ($p < 0.001$). All running at **\$0/month**.



What's Next

- Load **100% of IRS data** (now at **31%**)
- Add Redis for persistent caching
- Add historical FEC data back to **2016**
- Implement proper user authentication
- Expand analytics dashboard from **37% to 100%**

Capability Comparison

Capability	Investigative RAG (Ours)	Pure Vector RAG	SQL Only	ChatGPT / Generic LLM
Structured Data Query	Hybrid SQL-first routing	Imprecise fallback	Requires manual setup	Not specialized
Document Semantic Search	Good (Pinecone vectors)	Depends on precision	None	Good (but no citations)
Real Government Data	IRS + FEC live records	Depends on ingestion	Yes	Training cutoff – stale
Cross-Dataset Joins	SQL JOIN across IRS/FEC	Not native	SQL JOIN	Not possible
Cited Answers	Always cited (1.7M+)	Partial	Few rows only	Sometimes – not grounded
Response Speed	4.74s / 0.0s cached	5–15s typical	<1s	3–30s – unpredictable
Free to Deploy	Yes (all free tier)	Partially	Yes	No (paid API required)

Conclusion : Our hybrid approach achieves 96.8% accuracy — an 11.8% improvement over pure vector search (85%) and 6.8% over pure SQL (90%) — confirmed statistically significant at $p < 0.001$, making it the most accurate and complete solution for investigative financial Q&A.

Lessons Learned & Future Improvements

Lessons Learned:

- SQL-first routing beats pure RAG: We initially collected 100K+ embeddings, building to handle complex financial queries. SQL-first routing proved faster, cheaper, and more accurate.
- Free-tier constraints drive innovation: Supabase 500 MB cap forced us to load 31% of IRS records. Smarter caching and indexing as a workaround.
- Ground truth evaluation is non-negotiable: Without 125 hand-curated questions, we'd have missed the LLM-as-judge DeepEval framework with arc.
- Agent separation improves maintainability: Separating Filter, Retriever and Writer agents made debugging build → agent has one job and logs clearly.

Future Improvements

- Expand IRS dataset coverage beyond the 31% (100K+ records to full 1.2M+ by 2025).
- Persistent Redis Cache: Replace in-memory dict with Redis to cache survives server restarts and scales across instances.
- User Authentication & Rate Limiting: Add API key management to enable grade access control.
- Multilingual Query Support: Extend beyond English to Spanish, Mandarin, and French for international researchers.

Thank You

Contact & Links

- Live App: <https://capstone-group2-investigative-rag.vercel.app/>
- Git Hub: <https://github.com/saimanikantabattula/capstone-group2-investigative-rag>

